

YEAR 7 · SCIENCE

Q1 Review: Science Units 1 and 2

Science 1.1–1.4 — cells, specialised cells, tissues and organs

Science 2.1–2.4 — particles, changes of state, the water cycle

Science 2.5–2.8 — atoms, compounds, mixtures, acids and bases

Name: _____

Date given: _____

How to use this packet

Focus the basics · **Practice** like the assessment · **Challenge** the hardest — do not skip
Answer in full sentences. “Nucleus” is not a sentence.

Section A · Unit 1 — Cells, Tissues and Organs

A1 Every part has a job

FOCUS

Word help — two parts both “control” something

controls — is in charge of. **releases** — lets out. Read the words **after** “controls”: *what goes in and out* is one part; *everything the cell does* is another.

(a) Draw **one** line from each part of a cell to its job.

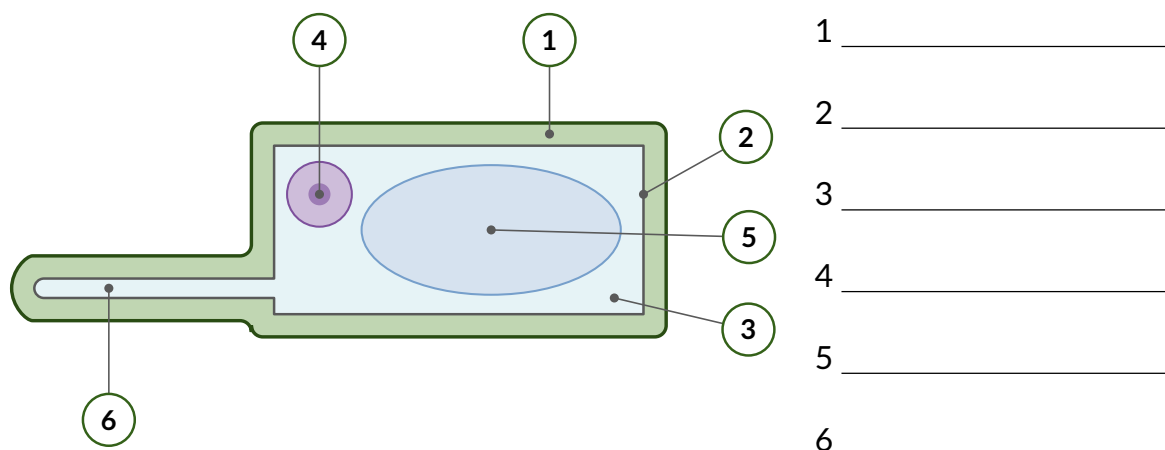
- | | |
|-----------------|---|
| cell membrane • | • makes food using sunlight |
| cytoplasm • | • releases energy from food |
| nucleus • | • holds a plant cell in shape |
| mitochondrion • | • controls what goes in and out of the cell |
| cell wall • | • is full of cell sap and keeps the cell firm |
| chloroplast • | • controls everything the cell does |
| sap vacuole • | • is where the cell's chemical reactions happen |

(b) Draw a ring around the **three** parts in (a) that an **animal** cell does **not** have.

A2 The root hair cell

PRACTICE

(a) Write the name of each numbered part.



(b) Water from the soil goes into the cell. Number these **1 to 4** in the order the water passes through them.

☐ cell membrane ☐ sap vacuole ☐ cell wall ☐ cytoplasm

(c) Why does this cell have a long, thin root hair?

The long root hair gives the cell ...

A3 Which level is it?

FOCUS

Each level is built out of the one before it

cell → **tissue** (similar cells, one job) → **organ** (several tissues) → **organ system** (organs, one shared job) → **organism**

Write each one in the correct column.

a neurone · the brain · ciliated epithelium · a mango tree · the digestive system
a root · the palisade layer · a red blood cell · the breathing system · a cat

Cell	Tissue	Organ	Organ system	Organism

A4 Mr Bowen's table has four mistakes**PRACTICE**

There is **one** mistake in each row. Cross it out and write the correct words next to it.

Cell	Function	Special structure
red blood cell	carries oxygen	has a very long axon
neurone	carries electrical signals	has a large sap vacuole
ciliated cell	absorbs water from the soil	has cilia (tiny hairs)
palisade cell	makes food by photosynthesis	has no nucleus

A5 Unit 1 puzzles**CHALLENGE****Word help**

side by side — in a line, touching. **squeezes** — pushes through a small space.

1. A typical human cell is 0.02 mm across.

(a) How many cells, side by side, make a line 1 mm long? _____

(b) How many make a line 1 cm long? _____

(c) A hair is 0.07 mm thick. About how many cells wide is it? _____

2. A white blood cell chases germs, squeezes between other cells and swallows the germs. Why would a cell wall be **no good** to it?

Section B · Unit 2 — Particles, Changes of State, the Water Cycle**B1 Draw the particles****FOCUS**

Draw about 12 particles in each box. Use circles of the same size.

**solid****liquid****gas**

B2 Solid, liquid or gas?

FOCUS

Tick every state it is true for. Some rows need two ticks.	Solid	Liquid	Gas
It keeps its own shape.			
It fills any closed container.			
It takes the shape of its container, but keeps the same volume.			
It can be poured.			
It cannot be compressed.			
The particles vibrate, but stay in the same place.			
The particles touch, but can move past one another.			

B3 Name the change

PRACTICE

Five changes of state

melting solid → liquid · **freezing** liquid → solid · **evaporating** and **boiling** liquid → gas
 · **condensing** gas → liquid

What happens	Change of state	Heating or cooling?
A bar of chocolate goes soft in a hot car.		
Mr Bowen breathes on a cold window. Drops of water appear.		
Melted butter goes hard in the fridge.		
A wet floor dries in the afternoon.		
Water in a kettle bubbles at 100 °C.		

B4 Explain it with particles

PRACTICE

(a) What happens to the particles when water boils?

Heat energy is transferred to ... The particles move ... Some of them escape ...

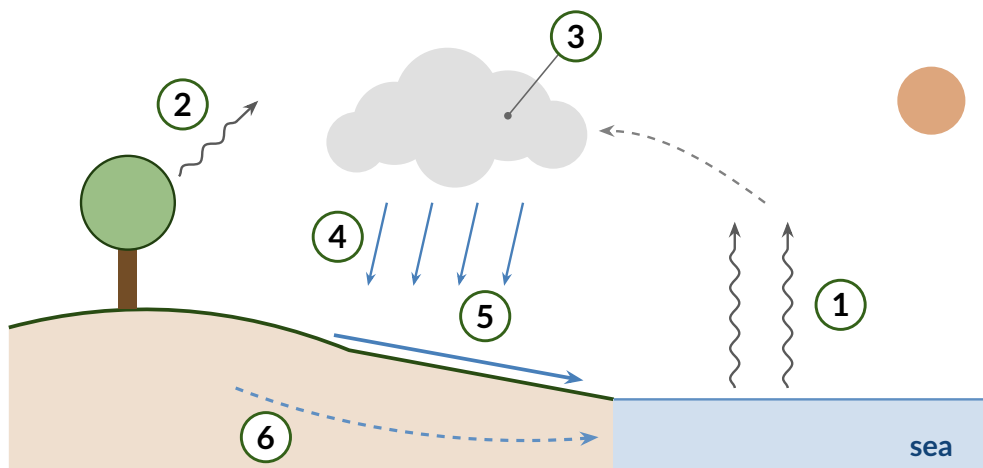
(b) Drops of water form on the outside of a cold can of drink. Where do they come from?
There is water vapour in ... When it touches the cold can, the particles ...

B5 The water cycle

PRACTICE

(a) Write the name of each numbered arrow. Use the words in the box.

condensation	evaporation	groundwater
precipitation	surface run-off	transpiration

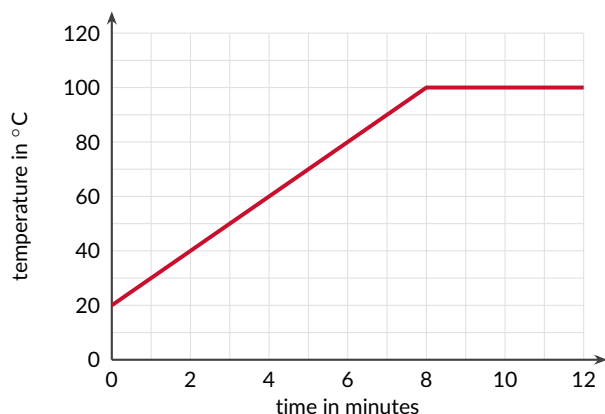


1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

(b) You can never see water vapour. So what is a cloud made of?

(c) Rain is one kind of precipitation. Name **two** other kinds. _____

B6 Particle puzzles**CHALLENGE**

1. Mr Bowen heats a beaker of water for 12 minutes. The graph shows its temperature.

(a) When does the water start to boil?

(b) He is still heating it at 10 minutes. What is the temperature?

(c) Why does the temperature stop going up, even though he is still heating it?

2. Wet clothes dry faster on a hot, sunny day than on a cold day. Explain why, using particles.

Section C · Unit 2 – Atoms, Compounds, Mixtures, Acids and Bases

C1 Symbols, both ways**FOCUS**

The capital-letter rule

The first letter is **always a capital**. A second letter is **always small**: Mg, not MG or mg. A few come from Latin names: potassium is **K**.

(a) Write the symbol.

oxygen _____ potassium _____

magnesium _____ calcium _____

nitrogen _____ aluminium _____

(b) Write the name.

He _____ S _____

Li _____ Ar _____

F _____ Si _____

C2 Rows and columns

PRACTICE

	group							
	1	2	3	4	5	6	7	8
period 1				H				He
2	Li	Be	B	C	N	O	F	Ne
3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca						

 = metal

period = a **row** (across), not a lesson
group = a **column** (down), not people

(a) Period 3, group 2: _____

(b) Calcium is in period _____

(c) Two in group 8: _____

(d) Li and Na are in the same _____

C3 Symbols and formulae

PRACTICE

Reading a formula

The small number counts the atom **just before it**. No number means **one**. CO₂ is 1 carbon + 2 oxygen = 3 atoms.

K	Cl ₂	MgO	CH ₄	Ca	H ₂ O	NaOH	K ₂ CO ₃
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Choose from the box.

(a) Write **two** elements made of single atoms. _____(b) Write the element whose particles are **two atoms** joined together. _____(c) Write the compound that is **cooking gas**. _____(d) Write the substance that contains a metal, carbon **and** oxygen, and give its name.
 _____ name: _____(e) How many atoms are in NaOH? _____ in K₂CO₃? _____

C4 Naming compounds

PRACTICE

Two naming rules

The **metal comes first**, and the non-metal ends in **-ide**: sodium + chlorine → sodium **chloride**. Two elements **plus oxygen** often end in **-ate**: calcium carbonate.

(a) Name the compound made from:

1. sodium and sulfur _____ 3. lithium and chlorine _____

2. oxygen and calcium _____ 4. magnesium, carbon and oxygen _____

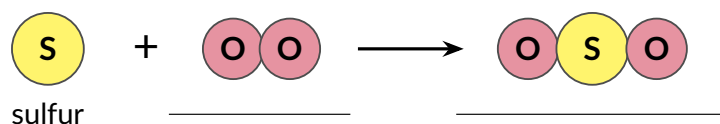
C7 Two questions from the end of the year

CHALLENGE

Word help

reactants — the substances you **start** with. **product** — the **new** substance that is made.
mean — add the results, then divide by how many there are.

1. Sulfur burns in oxygen. The picture shows the particles.



(a) Write the name of the other reactant, and of the product, under the picture.

(b) Write the formula of the product. _____

2. Mr Bowen has two bottles of alkali, **X** and **Y**. He puts 20 cm³ from one bottle in a flask with a few drops of universal indicator, and adds acid until it is neutralised. He does this three times for each bottle.

Bottle	Acid needed in cm ³			Mean
	1st try	2nd try	3rd try	
X	15	16	17	16
Y	23	26	23	

(a) How does he know when the alkali has been neutralised?

(b) Why does he do it three times, not once?

(c) Work out the mean for bottle **Y** and write it in the table.

(d) Which bottle holds the **stronger** alkali? Explain how you know.

Section D · Both Units

D1 Mr Bowen's lab notebook

CHALLENGE

Mr Bowen wrote this. It is full of mistakes. Find **six**: underline each one, then write the correct sentence.

Monday. I looked at my own cheek cells under the microscope. Each one had a cell wall and a big sap vacuole. Their mitochondria make food using sunlight.

Then I boiled some water for my tea. It boiled at 0 °C and turned into water vapour, which I could see as a white cloud above the kettle.

I put salt in my tea. Salt is a mixture of sodium and chlorine. I dipped red litmus paper in the salt water and it turned blue, so salt water is an acid. The tea was terrible.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____